

Research Note

Designing Promotion Ladders to Mitigate Turnover of IT Professionals

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Chronic excessive turnover among information technology (IT) professionals has been costly to firms for decades with annual turnover rates as high as 24% even among *Computerworld's* "100 Best Places to Work in IT." Prior information systems literature has identified two key factors affecting turnover: boundary-spanning roles and low promotability in one's current firm. We draw on tournament theory, which is primarily concerned with inducing effort in employees, to decompose promotability into two distinct constructs: the *likelihood of promotion* and *benefit from promotion*, and demonstrate that each has a distinct role in affecting turnover rates. Our key result is that a job ladder motivating IT professionals with large, infrequent promotions will lead to higher turnover than a job ladder with smaller, more frequent promotions. We describe the conditions under which rearranging the job ladder creates economic value for the firm. We also offer an explanation for the observation that jobs characterized by boundary-spanning activities have higher turnover, and show that such jobs are more sensitive to the effect of likelihood of promotion on turnover. We test our hypotheses on a detailed data set covering 5,704 IT professionals over a five-year period. We confirm that likelihood of promotion has the predicted effects on turnover of IT professionals. A one standard deviation increase in likelihood of promotion decreases turnover by over 99%, consistent with our prediction. The empirical analysis also confirms the predicted effects of boundary spanning activities.

Keywords: IT turnover; tournament theory; econometrics; career concerns; management of IT human resources

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1. Introduction

Chronic excessive turnover among information technology (IT) professionals has been costly to firms for decades (Jiang and Klein 2002, Tambe and Hitt 2014), during periods of high and low unemployment (Quan and Cha 2010) with annual turnover rates as high as 24% even among the "100 Best Places to Work in IT" (*Computerworld* 2015). Prior research (see Joseph et al. 2007 for a review) has identified several factors that impact turnover of IT professionals including job satisfaction, organizational commitment, human capital, intrinsic motivation, and external labor market conditions, with boundary-spanning activities and low promotability as key factors that are under the firm's control. Yet a firm cannot avoid boundary-spanning jobs, and increasing promotability can be costly to the firm. This leads us to the following questions: What trade-off does a firm face between promotion incentives, turnover, and overall

wage costs? Do the components of promotability affect IT professionals' incentives to remain with or leave their current employer? More specifically, what is the impact of likelihood of promotion and the benefit from promotion on the turnover rate? How do boundary-spanning activities affect turnover and why?

Our approach to these questions is to incorporate tournament theory into the information systems (IS) literature on turnover, and to our knowledge ours is the first study to do so. Tournament theory explains how relative performance appraisals incentivize effort from knowledge workers (Lazear and Rosen 1981, Rosen 1986), including IT professionals, even if the firm cannot observe effort levels. We combine prior research on tournament theory with economically significant job alternatives to generate novel recommendations for mitigating turnover among IT professionals.

Turnover often occurs when some job alternative's net present value (NPV) is sufficiently higher than the NPV of staying at the current job, particularly if the NPV of the current job has decreased (Lee et al. 1999).¹ This is related to two phenomena, "career concerns" (Gibbons and Murphy 1992, Gibbons and Waldman 1999) in which employees exert effort for rewards in the future, and "promotions as job market signals" (Waldman 1984, Lerner and Tirole 2002, Roberts et al. 2006, Owan 2004, Roberts et al. 2006), which extend the career horizon beyond the current employer. A promotion at the current firm may signal an employee's ability to other firms independent of signals that she can send herself such as education (Spence 1973). Firms may engage in a risky strategy of delaying promotion to mitigate poaching (Bernhardt 1995, Owan 2004), but on average promotions decrease turnover among IT personnel (Igbaria and Siegel 1992).

In the context of tournament theory with economically significant job alternatives, we move beyond correlation between pay dispersal and turnover (Bloom and Michel 2002) to discuss two key factors that have been identified in prior IS literature as impacting turnover: boundary-spanning activities and promotability.

Boundary-spanning activities are those that require interaction across multiple business functions or departments. They can sometimes be a source of power within the organization (Baroudi and Igbaria 1994) but are almost always a source of stress (Kahn et al. 1964), and many IT jobs fall into the boundary-spanning category (Baroudi 1985).

Promotability has been defined in prior literature as the probability of being promoted within one's current firm (Igbaria and Greenhaus 1992). Prior research has identified promotability within the current firm as a factor in turnover decisions (Igbaria and Greenhaus 1992, Igbaria and Siegel 1992, Nalbantian and Szostak 2004), but the variety of internal labor market strategies used by IT departments (Ang and Slaughter 2004) and the indeterminate results of these efforts (Joseph et al. 2007) indicate that promotability's effectiveness is still an open question.

A key contribution of our work is that promotability should be considered as two distinct constructs of *likelihood of promotion* and *benefit from the promotion*. After all, a promotion from Helpdesk Agent 2 to Helpdesk Agent 3 (with a small pay increase and similar prospects for promotion) is not as valuable as a promotion from senior developer to technical manager (with a large pay increase and a shift from a

career plateau to a highly promotable management role).

"[O]pportunities for career mobility outside the organization have been found to be negatively related to satisfaction and commitment," (Mobley et al. 1979; Bluedorn 1982; Cotton and Tuttle 1986; Igbaria and Greenhaus 1992, p. 37) which leads to turnover (Meyer et al. 2002, Paré and Tremblay 2007). The fungibility of IT skills and the need to acquire updated IT skills (Gallivan et al. 2002) increases turnover relative to other professions (Spain 1995, Ang and Slaughter 2004).

Our results show that incentives from internal promotion opportunities for IT professionals can have undesirable—but avoidable—effects on turnover. Our key result is that a job ladder motivating IT professionals with the possibility of large, infrequent promotions will lead to higher turnover than a job ladder with smaller, more frequent promotions. This is consistent with the reduced-form results in Johnston et al. (1993) and Bloom and Michel (2002), but our detailed microdata allow us to explore the conditions under which adjusting the job ladder is cost-effective for the firm. We argue that increasing the likelihood of promotion can reduce turnover without increasing the NPV of the firm's wage cost, although there is no free lunch: the firm faces a trade-off between incentive effects and turnover expenses. Firms in industries with low turnover costs will be better served by focusing their job ladder design to incentivize increased employee effort by having large, infrequent promotions. However firms in industries with high turnover costs should deploy a job ladder that reflects the value of retaining talent. We argue that high turnover costs should lead to a job ladder with smaller, more frequent promotions as compared to the job ladder used by firms in low-turnover-cost industries.

Counterintuitively, we also find that increasing the benefit from promotion can increase turnover when it is coupled with a lower likelihood of promotion. This result is counter to prior career concern results that do not take turnover into account. We also find that jobs characterized by boundary-spanning activities are more sensitive to the effect of likelihood of promotion.

Our results have important implications for internal incentives that depart from classic tournament theory. Misapplying a ladder suitable for a low-turnover industry in a moderate-turnover industry creates precisely the type of situation that causes excessive turnover. Smaller, more frequent promotions reduce turnover even if the benefit from promotion is decreased so much that the NPV of the job is unchanged. Thus our work contributes to the literature by showing the existence of a force countering the incentives created by promotion opportunities.

¹ This mechanism has shown good predictive power in several contexts, although interestingly not for IT professionals (Niederman et al. 2007, Mourmant 2009).

2. Hypothesis Development

In this section, we motivate the development of our hypotheses by combining tournament theory (Rosen 1986) with the future-oriented nature of career concerns (Gibbons and Murphy 1992). Lazear and Rosen (1981) show that even if the actual effort exerted by each employee is unobservable, the publicly observable characteristics of a tournament-style job ladder determine the equilibrium level of effort and the probability of promotion. The ability to design the job ladder allows the firm to induce a specific level of employee effort chosen from a fairly broad range. We argue that the “optimal” level of effort in a job ladder that does not consider turnover is not necessarily profit maximizing once turnover-related costs are considered.

Faced with a trade-off between incentivizing effort and mitigating turnover, the firm has three fundamental options. We discuss each option in turn.

1. The benchmark case is to induce an “optimal” level of effort ignoring the cost of turnover.
2. Increase employee compensation to make outside job alternatives less attractive.
3. Minimize the pool of at-risk employees and/or decrease the shock of nonpromotion.

The first option may be appropriate if the firm’s replacement cost is very low for this job. In this case, the cost of reorganizing the job ladder may not pay off for the firm (Cappelli 2000). Some entry-level IT jobs may be in this category such as helpdesk agents, but most IT staff are valuable enough to the firm to justify some expenditures on retention.

The second option is appropriate if the firm needs to adjust to labor market conditions that change rapidly relative to the firm’s ability to adjust its job ladder. Firms that enjoy a productivity advantage because of firm-specific human capital are in an especially good position to share some of their surplus with employees to reduce turnover.

The third option allows for compensation adjustments like option 2 but provides additional flexibility regarding the job ladder. For example, the firm can insert new levels in the middle of the job ladder that increase the fraction of employees promoted but decrease the incremental benefit of each promotion. Such an adjustment could be less costly, but it would also weaken employees’ incentives for exerting unobservable effort. This option is appropriate if the firm has time to adjust its job ladder such as when it is facing persistent, long-term turnover issues. We devise our hypotheses in a way that controls for firm actions consistent with the second option, so that we may focus on the third option, which is novel to our research.

We expect that employees will seek to maximize their NPV over their entire careers derived

from wages less the cost of effort and nonpecuniary costs associated with switching employers, plus any nonpecuniary benefits from working at the firm (Duncan 1976). The firm’s NPV, on the other hand, is the employee’s productivity less the NPV of wages and the cost of nonpecuniary benefits. Consider an employee’s NPV just prior to an appraisal for a possible promotion. Her NPV is a weighted average of the NPV conditional on promotion and the NPV conditional on nonpromotion. If the employee is promoted then her NPV increases, otherwise she is not promoted and the NPV decreases. Nonpromoted IT professionals have job alternatives that may exceed the NPV of remaining until the next appraisal, causing some to leave the firm. Promoted employees may also leave the firm because of job market signaling, but they do so at a lower rate (Igarria and Siegel 1992, Johnston et al. 1993).²

Ceteris paribus, increasing the promotion rate will put more employees into the promoted group that has lower turnover, lowering the overall turnover rate, but increase costs for the firm. One way to increase promotion probability at lower cost is to redesign the job ladder with more rungs. The additional rungs will slow down the increase in the firm’s costs, and it can even be done in a way to keep the NPV of costs constant while increasing promotion probability. However, employees are motivated by the difference in NPV between promotion and nonpromotion (Lazear and Rosen 1981), and increasing promotion probability without increasing the overall NPV must decrease the difference in NPV between promotion and nonpromotion. This smaller difference will induce less unobservable effort, creating a trade-off between promotion incentives and turnover mitigation. Hence tournament theory incentivizes effort but also introduces shocks, which lower NPVs whenever an appraisal does not result in a promotion. Thus, we have the following:

HYPOTHESIS 1. *Controlling for NPV, the turnover rate is lower for jobs with larger likelihood of earning a promotion.*

As the likelihood of promotion increases, the fraction of employees that are not promoted decreases. As discussed previously, the turnover rate is lower for promoted employees (Johnston et al. 1993), so this leads to a reduction in turnover. Introducing the NPV of holding the job as a control variable removes the effect of changes in the average wage (i.e., any use of the “second option” is filtered out). The firm can balance the wage cost implications of the increased

² Rational firms avoid promoting someone if doing so increases her *ex ante* probability of leaving the firm.

likelihood of promotion by reducing the benefit from promotion.

In Hypothesis 2, we test for the effect of benefit from promotion upon turnover, again controlling for NPV. It is possible that this control may not fully filter out the “second option.” In Hypothesis 2, any impact from the second option “seeping” through will decrease turnover. However, if the control works properly, then the third option will dominate and turnover will increase because of the larger drop in employee NPV. Therefore, this hypothesis tests the adequacy of our control and our novel prediction that an increase in the benefit from promotion will lead to an increase in turnover.

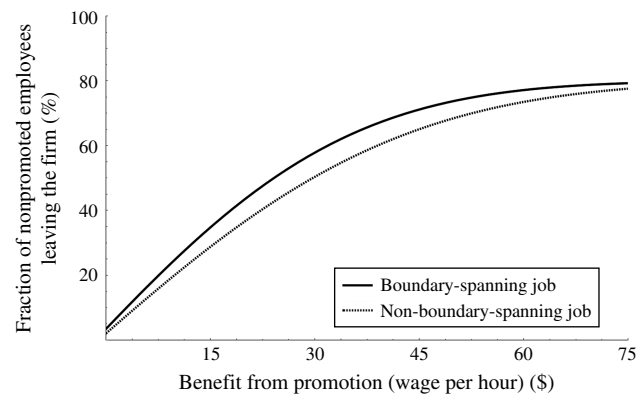
As discussed above, an appraisal that does not result in a promotion creates a negative shock to the employee’s NPV. The higher the potential benefit from the promotion, the larger the negative shock and the greater the likelihood of the nonpromoted employee leaving the firm. If our prediction about the “third option” is true, then a higher benefit from promotion will tend to occur with a lower likelihood of promotion, increasing the pool of employees that are subjected to this negative shock. Thus, we have the following:

HYPOTHESIS 2. *Controlling for NPV, jobs with higher benefit of promotion will have higher turnover.*

Our first two hypotheses assume implicitly that the only things that differentiate jobs are objective measures like NPV, *Likelihood of Promotion*, *Benefit from Promotion*, and *Current Wages*, but this obviously oversimplifies the problem. Some jobs are associated with lower commitment to the firm than others (George and Bettenhausen 1990), which has important implications for turnover. It turns out that many IT jobs involve boundary-spanning activities that increase stress (Kahn et al. 1964) and decrease job satisfaction and organizational commitment (Baroudi 1985, Guimaraes and Igbaria 1992) relative to other jobs.³ Hence, we expect boundary-spanning activities to be significant drivers of employees’ commitment to the firm. *Ceteris paribus* such activities would tend to increase turnover because on average these employees are less attached to their current employer, as stated in Hypothesis 3A and illustrated by how the solid curve (turnover rate for boundary-spanning jobs) is higher than the dashed curve (turnover rate for non-boundary-spanning jobs) in Figure 1.

³ The decrease in commitment might be due to the nature of the job making the employee more willing to leave, the boundary-spanning itself creating contacts that allow the employee to find job alternatives more easily, or some combination of the two. Once a job is designed to be boundary spanning, these commitment effects are largely outside the firm’s control.

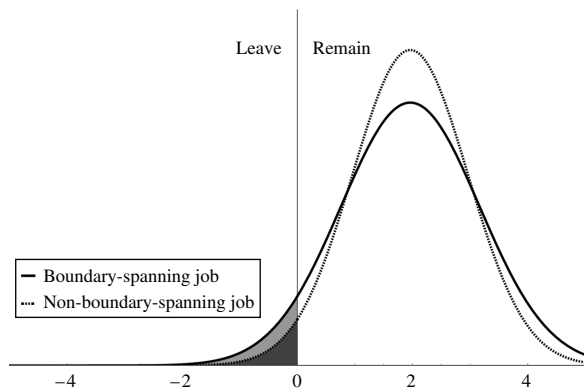
Figure 1 Effects of Boundary Spanning upon Turnover



Notes. Curve represents the cumulative fraction of nonpromoted employees who leave the firm for different levels of *Benefit from Promotion*. The solid curve represents boundary-spanning jobs, and the dashed curve represents non-boundary-spanning jobs. The numbers used in this figure are for illustration only.

In addition to having lower commitment on average, we expect that employees in boundary-spanning jobs will also exhibit greater turnover because of higher variance in their level of commitment than those in nonspanning jobs. Their variance is higher because employees in boundary-spanning IT jobs are subject to all of the factors that affect organizational commitment of IT employees and in addition are heterogeneous in (1) their tolerance for type of stress induced by boundary-spanning activities and (2) their valuation of the power that can be derived from a boundary-spanning role (Kahn et al. 1964, Tushman and Scanlan 1981). So long as turnover is less than 50%, higher variance means that employees with below-average commitment are more likely to fall below the threshold for actually leaving, as illustrated in Figure 2. Recording the fraction below the threshold for many values of *Benefit from Promotion* yields the original Figure 1, and note that the solid curve has a higher slope in the lower left of Figure 1 indicating that the outcome (turnover rate) is more sensitive to small changes in the variable (*Benefit from Promotion*). Since turnover rates for IT jobs are high but significantly less than 50%, we expect that the higher variance in commitment will cause small changes in *Benefit from Promotion* to have larger effects on them as stated in Hypothesis 3B.⁴

⁴ On the other hand, if these employees operate with minimal commitment to the firm then their likelihood of leaving could be “saturated,” and they would be less sensitive to differences in *Benefit from Promotion* as in the upper right part of Figure 1. The symptoms of extremely low commitment (pervasive shirking, absenteeism, turnover, etc.) would prompt management to redesign the job, so “saturation” is unlikely to be a long-run feature of any job in the ladder.

Figure 2 Variance in Commitment

Notes. These distributions have the same mean and support, but the solid one has greater variance. Only 2.5% of the dashed distribution falls below the threshold to leave the firm (thin vertical line), but 5% of the solid distribution falls below the same threshold. The numbers used in this figure are for illustration only.

HYPOTHESIS 3A. *On average, jobs characterized by boundary-spanning activities will have higher turnover than other jobs.*

HYPOTHESIS 3B. *The turnover effect of benefits from promotions identified in Hypothesis 2 will be greater in jobs characterized by boundary-spanning activities compared to jobs that are not.*

To test that our hypotheses describe economically significant factors, we perform a series of empirical tests in the next section.

3. Empirical Investigation

The focal firm in our study is a large multinational outsourcing, systems integration and consulting firm. It provides application managed services, IT consulting services, and numerous software solutions. The firm is headquartered in North America with offices in over 100 cities across four continents. They have over 30,000 employees. For this study, we based all employee-level analysis on the firm's administrative records to preserve the information asymmetries present in the workplace and prevent a Hawthorne Effect.⁵ As a fortunate side effect, this eases replication of the results across firms since most firms keep similar records.

Our primary data source is the firm's human resources records, which includes demographic information and date of hire for each employee. We also observe time-varying job titles, base salaries, and scheduled hours for each employee. We supplement this data with information from timecards that allow us to observe exits from the firm. These observations

Table 1 Summary Statistics of Job-Week Variables ($N = 3,252$)

Variable	Mean	Std. dev.	Min.	Max.
Wage per hour ($\bar{w}$)	1.000	0.493	0.115	2.674
Fraction part-time (%)	2.699	12.511	0.000	100.000
Turnover rate per week (%)	0.224	2.494	0.000	100.000
Likelihood of promotion ^a (%)	1.568	2.913	0.051	20.00
Benefit from promotion ($\bar{w}$)	0.307	0.218	1.906×10^{-5}	1.404
Job type dummy: Clerical	0.223	0.416	0	1
Job type dummy: CSR	0.138	0.345	0	1
Job type dummy: Developer	0.144	0.351	0	1
Job type dummy: Consultant	0.319	0.466	0	1
Job type dummy: Manager	0.176	0.381	0	1

^aWithin the next quarter. This variable is constant for a job across the entire time period, because the average promotion rate is the best proxy available for what the optimal promotion rate would be absent managers' endogenous attempts to retain specific employees.

cover 5,704 individual knowledge workers in 28 jobs ranging from customer service representatives to vice presidents over a five-year period from February 2005 to April 2010. Our data set does not include all of the firm's employees, but it does have full coverage of the employees for several business units.

We aggregate these data into 3,252 job-week observations using simple means (each person-week observation in the same job is averaged with equal weights). Summary statistics for these variables are presented in Table 1. There are two variables however, *Wage per Hour* and *Benefit from Promotion*, that require further explanation. *Wage per Hour* is converted from the local currency into nominal U.S. dollars based on the exchange rate recorded for Friday of that week, and to maintain privacy we report wages in units of the mean wage $\bar{W}$. Ideally our wage measure should also account for the cost of effort and nonpecuniary benefits in each job. The proxies for effort in our data (hours worked and absenteeism) are consistent across jobs, so the cost of effort falls out of the equation.⁶ We do not attempt to monetize the nonpecuniary benefits of the jobs within our data set, instead using categorical variables for job types explained in Table 2.

Operationalizing *Benefit from Promotion* is somewhat complex. We begin with a transition matrix of observed job changes within the firm in Table 3, in

⁵ Polling employees weekly about their intent to leave the company might affect their decision processes.

⁶ This consistency is almost certainly an artifact of data collection (for example, we have anecdotal evidence that managers participate in international conference calls late at night without billing that time), but unobserved higher cost of effort in higher jobs would bias against us finding results.

which we use observed transitions to define hierarchical levels rather than job titles or pay ranges (Baker et al. 1994). These levels are indicated on the matrix by a heavy box surrounding cells in the same hierarchical level, and shaded cells represent a move up one level. The matrix's row headings represent the job an employee performs at the start of a quarter (with job titles disguised to maintain confidentiality), and column headings represent jobs at the start of the next quarter. The number in each cell is the percentage of employees who started the quarter in the row's job who transitioned to the column's job. For every job, the most common outcome is to remain in the same job. The [Entry] row represents new hires during the sample period, while the [Exit] column represents separations from the firm. Using this matrix, we calculate the NPV of wages associated with occupying a job by incorporating the likelihood of promotion in the near future.⁷

Next, for jobs below the top level (level 8), we identify the "goal" for each job as the most common promotion earned by employees in that job.⁸ For example, the goal for a "Consultant 4a" is to become a "Consultant 5a." Finally, we define *Benefit from Promotion* as the NPV of the goal job minus the NPV of the current job. This variable is measured in dollars per hour and is one of the key variables of interest in our hypotheses.

We also coded the job titles in Table 2 according to the relative level of boundary-spanning activities in each job. Note that the suffixes in Table 3 do not indicate within-level steps on the job ladder, so each category in Table 3 represents a number of distinct jobs.

Although we wish to study effects at the job level to inform job ladder design, we must first validate that our variables of interest actually influence employee behavior. Table 4 shows that employees facing higher *Likelihood of Promotion* or lower *Benefit from Promotion* are less likely to leave the firm in a given week.

We use a Tobit estimator for our main empirical model to test our hypotheses because the dependent variable is limited (it cannot be below 0% nor above 100%) and ordinary least squares regression is biased whenever a significant number of observations are at the limits (which is the case with our data). The Tobit estimator is able to handle these limit cases if the

⁷ Employees are generally considered for promotion once per year during annual performance appraisals. We do not leverage the expectation of annual appraisals because there is not enough information in the data to infer everyone's appraisal schedule, and we do observe out-of-cycle job changes.

⁸ Data on level 8 employees determine the promotion opportunities for level 7 employees, but we cannot measure the incentives facing the level 8 employees themselves. Therefore, vice presidents are excluded from the analysis.

Table 2 Job Types in the Focal Firm

Job type	Brief description
Clerical	Non-IT jobs in the IT hierarchy: secretary, underwriter, etc.
Consultant	High-skill IT jobs that are customer facing: business analyst, team leader, etc.
Customer service representative	Low-skill IT jobs that are customer facing: Help desk, sales support, etc.
Developer	IT jobs that are not customer facing and thus less boundary spanning than typical IT jobs: database administrator, programmer, etc.
Manager	Managerial jobs that routinely interact with IT personnel (either as direct superiors or as the superiors of clerical employees): director of consulting, technical manager, etc.
Subcontractor	These jobs are excluded from our analysis because their job title never changes. They do not appear in Table 3.

data meet certain assumptions, which we are able to accommodate. Our empirical specification is

$$\begin{aligned}
 \text{turnover}_{j,t}^* &= \beta_0 + \beta_1 \text{promoBenefit}_{j,t} + \beta_2 \text{promoLikelihood}_{j,t} \\
 &+ \beta_3 \text{NPV}_{j,t} + \beta_4 \text{partTime}_{j,t} + \sum_{k=1}^4 \gamma_k \text{jobType}_{k,j} \\
 &+ \sum_{k=1}^4 \delta_k (\text{promoBenefit}_{j,t} \times \text{jobType}_{k,j}) + \varepsilon_{j,t}, \quad (1)
 \end{aligned}$$

where

$$\text{turnover}_{j,t} = \min[1, \max(0, \text{turnover}_{j,t}^*)], \quad (2)$$

and the *promoBenefit*_{*j,t*} variable is mean centered so that it can be interacted with other variables without biasing the results. We calculate our monetary variables using three regimes: without discounting, with adjustments for a fixed discount factor, and with adjustments for inflation.

The dummy variable *jobType*_{*k,j*} is one if job *j* is in category *k* and zero otherwise (*k* = 0 represents the excluded category, Clerical). The variable *turnover*_{*j,t*}^{*} represents intended or latent turnover while *turnover*_{*j,t*} represents observed turnover (which cannot be below 0% nor above 100%). The NPV of occupying the job and the fraction of employees in each job who work part-time are included as control variables.

Our empirical results for the above model are presented in Table 5 using nominal dollars and

Table 3 Job Transition Matrix

[illegible]

Table 4 Estimation Results for Individual Turnover ($N = 404,015$)

	Panel logistic regression (population average)
Dependent variable	Exit during focal week
Observation	Person-week
Likelihood of promotion	–278.045** (138.745)
Benefit from promotion	0.037*** (0.005)
NPV	–0.014** (0.005)
Part-time employee	0.629*** (0.207)
Female	0.324*** (0.077)
Age at hire (years)	–0.013*** (0.004)
Tenure in firm (years)	–0.115*** (0.012)
Tenure in current job (years)	0.088*** (0.013)
Intercept	–4.740*** (0.294)

Note. Standard errors are in parentheses.

** $p < 0.05$; *** $p < 0.01$.

Table 6 deflated to December 2006 dollars.⁹ Using raw fractions as our turnover measure did not meet the normality assumptions of the Tobit estimator, so our model uses a logit transform of the same fraction as the dependent variable, and in this specification the normality assumption is satisfied. The coefficients in these tables are analogous to the coefficients of a binomial logit estimator in that they estimate how our variables of interest change the probability of observing an exit for a randomly chosen individual in that job.¹⁰

Although the “Fisher effect” postulates that exchange rates account for differences in inflation in expectation, in practice the response is slow (Mishkin 1992) and therefore the use of inflation-adjusted dollars introduces considerable instability into the estimates. Therefore, Table 5 is our preferred specification despite Table 6’s higher R^2 scores. Those higher R^2 scores do, however, indicate that employees seem to rationally anticipate the effects of inflation.

In Table 5, Model (a) includes *Likelihood of Promotion* plus controls, Model (b) includes *Benefit from Promotion* plus the same controls, and Model (c) adds interaction terms between mean-centered *Benefit* and the job categories. For the specification shown in Table 5, tests for multicollinearity showed that

the highest variance inflation factors for our models are 4.01, 2.85, and 8.76, respectively, which are lower than the traditional limit of 10. Since estimation errors of the same job may be correlated over different weeks, we also employed alternate specifications using panel estimators at the job level in the online appendix (available as supplemental material at <http://dx.doi.org/10.1287/isre.2016.0652>).

We test Hypothesis 1 using the variable *Likelihood of Promotion* in Table 5. A higher expected *Likelihood of Promotion* is associated with lower *Turnover*. Model (a) shows that at the mean turnover rate, a one standard deviation increase in *Likelihood of Promotion* could virtually eliminate *Turnover* by reducing it to $(1.676 \times 10^{-4})\%$.¹¹ This supports Hypothesis 1.

In Table 5’s Model (b), a higher *Benefit* is associated with higher *Turnover*. At the mean turnover rate of 0.224% per week, a one standard deviation increase in *Benefit* could more than quadruple *Turnover* to 0.936%, which is a statistically and economically significant change. Therefore Hypothesis 2 is supported, and our focal firm’s behavior is consistent with our reasoning for the third option in Section 2.

Next, Models (a) and (b) show partial support for Hypothesis 3A, in that the boundary-spanning jobs of Consultant and Manager have significantly higher turnover than Clerical employees (the excluded group). This is only partial support because boundary-spanning Customer Service Representatives do not have higher turnover while the Developers, whom we coded as not boundary spanning, do.

Finally, we test Hypothesis 3B in Model (c) using interaction terms; now it is evident that *Benefit* has a significantly increased effect on the boundary-spanning jobs of Consultant, Customer Service Representative, and Manager compared to the nonspanning jobs of Clerical and Developer. Although the main effect of *Benefit* is negative, the significant coefficients on the interactions are larger than the main effect resulting in net positive influence for *Benefit* on *Turnover*.

Taken together, our results provide evidence that promotability, boundary-spanning activities, and persistent job alternatives have economically significant impacts on the turnover of IT professionals. A higher likelihood of promotion is associated with lower turnover while a higher benefit from promotion is associated with higher turnover. Both of these effects match our predictions, both are absent from classic tournament theory, yet both fit with prior correlational studies of IT personnel turnover

⁹ We omit the results using a fixed discount rate because they have very similar coefficients and precisely the same pseudo- R^2 scores as Table 5.

¹⁰ “Randomly chosen” is critical here. Assuming that this aggregate probability of exit applies to any specific individual would be an ecological fallacy.

¹¹ This value is lower than the lowest nonzero turnover rate in our data set. Of course, it is also very close to the many zeros we observe.

Table 5 Estimation Results for Turnover ($N = 3,252$)

	Model (a)	Model (b)	Model (c)
Estimation method		Tobit with robust standard errors	
Dependent variable		Logit of turnover rate ^a	
Observation		Job-week	
Likelihood of promotion	−168.132*** (41.911)		
Benefit from promotion		0.184*** (0.028)	−0.296* (0.176)
NPV	−0.175*** (0.025)	−0.113*** (0.017)	−0.151*** (0.023)
Fraction part-time	0.212 (1.340)	2.907*** (1.388)	−0.410 (1.277)
Job: Consultant (spanning)	9.135*** (0.889)	9.353*** (0.840)	10.279*** (1.048)
Job: CSR (spanning)	−0.567 (1.422)	0.203 (1.392)	−0.025 (1.808)
Job: Developer (nonspanning)	6.102*** (0.876)	7.055*** (0.912)	7.029*** (1.141)
Job: Manager (spanning)	9.663*** (1.151)	7.528*** (0.976)	7.429*** (1.395)
<i>Benefit × Consultant</i> (spanning)			0.322* (0.181)
<i>Benefit × CSR</i> (spanning)			0.901** (0.439)
<i>Benefit × Developer</i> (nonspanning)			−0.138** (0.227)
<i>Benefit × Manager</i> (spanning)			0.636*** (0.181)
Intercept	−14.533*** (1.051)	−19.768*** (1.057)	−17.248*** (1.138)
Pseudo- R^2	0.086	0.084	0.102

Notes. Standard errors are in parentheses. Excluded category for Job is Clerical (Nonspanning).

^aThe lower limit is −8, indicating no exits for that role-week. The lowest nonlimit result is −7.579.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

(Mourmant 2009). Additionally, IT jobs characterized by boundary-spanning activities exhibit higher turnover and are even more sensitive to the turnover-inducing effects of benefit from promotion.

However, the *Benefit*-related coefficients we observe could be caused by an alternate mechanism: if the employees who exit the firm happen to be those earning the highest wages in their respective jobs (perhaps because the job alternatives are easier to obtain for such employees), such exits would pull down the average wage for the focal job without affecting the average wage of the goal job, and artificially increase the observed *Benefit*. Yet we can exclude this possibility for two reasons. First, examination of the data reveals that the 12% of employees exiting the firm each year tend to earn wages in the 40th to 50th percentile for their jobs at the focal firm. Second, we designed our NPV variable to include the exiting employee's wages in the average wage to better represent the *Benefit* facing the employees in that job.

Another potential concern for our results' validity is endogeneity of a sort that managers intentionally promote employees that they believe are most likely

to leave the firm (using signals unobserved by the researchers). This could be serious because we do not have access to a good instrument for which employees should be promoted absent their managers' discretion. The best proxy available is the average promotion rate for that job over the entire sample period, which is the control we use. If we insufficiently control for managers' preferential promoting, the effect would be to depress the turnover rate for nonpromoted employees and bias *against* us finding results for *Likelihood of Promotion* or boundary spanning.

4. Discussion and Conclusion

We extend tournament theory with economically important job alternatives to explain the turnover behavior of IT professionals. In this section, we provide a summary of our key findings, contributions to the literature, and managerial implications of our results.

Our results show that the "promotability" construct from prior literature has two distinct components that impact turnover at the job level in a promotion tournament environment. First, the *Likelihood of*

Table 6 Inflation-Adjusted Estimation Results for Turnover ($N = 3,252$)

	Model (d)	Model (e)	Model (f)
Estimation method		Tobit with robust standard errors	
Dependent variable		Logit of turnover rate ^a	
Observation		Job-week	
Likelihood of promotion	−198.794*** (48.864)		
Real benefit from promotion		0.166*** (0.031)	−63.726* (36.592)
Real NPV	−0.233*** (0.028)	−0.150*** (0.020)	−0.251*** (0.042)
Fraction part-time	−0.522 (1.426)	2.404** (1.116)	−1.095 (1.286)
Job: Consultant (spanning)	10.015*** (0.921)	9.853*** (0.825)	313.883 (421.260)
Job: CSR (spanning)	−0.811 (1.425)	0.048 (1.382)	1,850.132** (753.770)
Job: Developer (nonspanning)	6.419*** (0.870)	7.311*** (0.899)	−357.748 (511.827)
Job: Manager (spanning)	11.315*** (1.215)	8.962*** (0.984)	1,561.039*** (400.262)
<i>Benefit × Consultant</i> (spanning)			28.668 (39.911)
<i>Benefit × CSR</i> (spanning)			175.477** (71.504)
<i>Benefit × Developer</i> (nonspanning)			−34.677 (48.462)
<i>Benefit × Manager</i> (spanning)			147.067*** (37.893)
Intercept	−12.958*** (1.115)	−18.772*** (1.081)	−687.329* (386.480)
Pseudo- R^2	0.100	0.089	0.114

Notes. Standard errors are in parentheses. Excluded category for Job is Clerical (Nonspanning).

^aThe lower limit is −8, indicating no exits for that role-week. The lowest nonlimit result is −7.579.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Promotion in each period is an important driver of turnover because promoted employees are less likely to leave the firm, and obviously a higher *Likelihood of Promotion* puts more of the employees in the promoted group. Second, the *Benefit from Promotion* is highly correlated with the negative shock experienced by employees who are not promoted,¹² with larger shocks causing a larger fraction of the nonpromoted employees to leave the firm. Taken together, these results demonstrate why plateau-type jobs that offer a large but less likely path to promotion lead to consistently high turnover. It is important to note that the mechanisms we describe apply to any knowledge workers facing competitive promotions, but the high baseline turnover in the IT profession makes these mechanisms particularly salient because small changes can have economically important impacts for the firm.

These mechanisms are even more salient for the subset of IT jobs that involve considerable

boundary-spanning activities. We find partial support for our hypothesis that these jobs have higher turnover on average, and we find full support for our hypothesis that boundary-spanning IT employees are more sensitive to the *Benefit from Promotion* effect. If small changes are important for IT jobs in general, even the smallest adjustment to boundary-spanning IT jobs can have an outsized impact.

For example, consider the customer service representative jobs at level 2 of our focal firm's hierarchy (see Table 3). The promotions for a CSR 2a or CSR 2c have large benefits (to Developer 4 or Consultant 4a) but relatively low likelihood. The promotion for a CSR 2b is more modest (to Clerical 3e) but at a much higher likelihood. We hypothesized that the CSR 2b job would have a lower turnover rate than CSR 2a or CSR 2c, and it does.

The intuition behind this result is as follows: just before they are appraised, each CSR's NPV is a convex combination of the values of her current job and the one for which she is competing. Each also has access to a job alternative with its own NPV. The CSRs

¹² For hyperirrational *Homo Economicus* employees, they would be perfectly correlated.

that are promoted now have a higher NPV for remaining at the firm (namely, the NPV of the new job) and are less likely to leave for a job alternative. The CSRs who are not promoted now have a lower NPV (because they must wait another period for the next appraisal and promotion opportunity) and are more likely to leave the firm for a job alternative. Since the likelihood of promotion is higher and the benefit of promotion is smaller for CSR 2b, this job has a smaller pool of high-risk employees *and* a smaller fraction of those employees leave the firm.

This finding, which applies much more broadly than just customer service representatives, has important managerial implications for the design of the job ladder. It suggests that when high turnover is a problem, firms can trade off between increased compensation, promotion incentives, and turnover. The firm can buy organizational commitment by paying an information rent to the employees (due to their private knowledge of their organizational commitment),¹³ but this can lead to wage inflation. This work provides another way to lower turnover: by increasing the number of rungs on the job ladder. This is profit improving so long as managers are careful to maintain enough promotion incentive to induce effort.

Counterintuitively, we also find that increasing the benefit from promotion can increase turnover. This result is counterintuitive because one would expect higher benefits to incentivize employees to stay at their current employer. This surprising result is, however, consistent with our hypotheses. We also predicted that IT jobs characterized by boundary-spanning activities would exhibit higher turnover. This result was partially supported by our empirical analysis, in that most job categories satisfied the hypothesis's prediction. We also predicted that such boundary spanning jobs would be even more sensitive to the turnover-inducing effects of benefits from promotion. This result was fully supported by our empirical analysis.

Prior literature has identified promotability as a crucial factor in IS turnover, and we contribute to this literature by examining jointly the likelihood of promotion and the benefit of promotion. Prior findings on the negative relationship between likelihood of promotion and turnover could have been driven by the increase in NPV that results from an increase in the likelihood of promotion. By considering the benefit from promotion, we are able to disentangle the cause of this correlation and determine that it is not simply the result of paying IT professionals more.

We also contribute to tournament theory by considering economically important job alternatives for the employees. IT professionals are an ideal population to demonstrate the importance of the job alternative's impact on tournament theory because IT professionals have persistent outside employment opportunities in a wide variety of economic conditions.

Since the incentives of a classic tournament affect not only effort decisions but also turnover, it is important for researchers applying tournament theory in future studies to account for the job alternatives when it is an economically significant factor for an industry (such as it is for IT professionals). Furthermore, for researchers studying turnover, it is important to take into account the various dimensions of a job's NPV identified by tournament theory when designing surveys or gathering administrative data. That is, while a single "promotability" survey item captures an important facet of promotion incentives, a richer set of items can capture the multidimensional impacts of internal promotion opportunities upon turnover.

Our results also have important implications for managers. Job ladders can be designed in a way that trades off incentive effects and turnover. Managers can accomplish this by increasing the likelihood of promotion at each appraisal (and balancing this with a lower benefit from the promotion to contain the overall wage cost). This can be accomplished either through marginal adjustments to existing jobs or by inserting new levels into the job ladder. Importantly, since boundary-spanning activities make IT professionals particularly sensitive to these types of shocks, even small changes can have economically significant impacts on turnover. Our study provides guidance on the constraints facing a manager designing (or altering) a job ladder for IT professionals.

On the other hand, if a firm wishes to decrease turnover without muting promotion incentives then it has the option to increase its compensation to its employees. This is consistent with SAS Institute's retention strategy of spending \$6,000 to \$8,000 per employee on perks to prevent \$50 to \$70 million of replacement costs (Harvard Business School 2002, pp. 80–81) and with the use of costly "industrial" internal labor market strategies to retain employees that would be expensive or disruptive to replace (Ang and Slaughter 2004). We believe that even a firm that is buying commitment from its employees (Spence 1973) can take advantage of our results to trade off a small effect on incentives to save a considerable amount of compensation.

Classic tournament theory's guidance to managers is to motivate employees by increasing the difference in present values between winning and not winning a promotion. Since classic tournament theory abstracts away turnover, the guidance may lead

¹³ Although their analysis has no turnover in equilibrium, Lazear and Rosen (1981) state that an off-equilibrium adjustment to raise wages is more efficient at retaining/poaching employees than increasing the likelihood or benefit of promotion.

to some steps of the job ladder with high benefits from promotion and low likelihood of promotion. We show that this combination increases turnover. One method for avoiding high-benefit/low-likelihood steps is to increase the number of rungs in the job ladder (without materially affecting the overall wage bill or the top and bottom wage levels) so that employees face smaller promotions with higher probabilities of attaining them.

Our study has limitations that researchers and managers may need to consider. First, we infer turnover from a premature exit from our panel data set, although such an exit may represent merely a transfer to a different line of work (e.g., a CSR 3 moving to sales). Such events create similar replacement costs for the focal business unit, but may reduce replacement costs or time-to-proficiency at the hiring business unit. Second, we cannot distinguish voluntary turnover from involuntary turnover, although from conversations with managers at the focal firm, it is apparent that firings are rare and there were no layoffs during the sample period. Voluntary and involuntary turnover impose similar replacement costs, but our results and guidance apply only to voluntary turnover. Third, some of the retention actions taken by the firm may trigger reactions from other firms. Fourth, the framework we test is an extension of tournament theory and is applicable only when tournament theory itself is applicable (costly or infeasible to observe individual effort or productivity, and an expectation that a significant number of employees will not be promoted in each period).

Future work in this area may include deriving the labor market equilibrium, which should demonstrate whether broad adoption of our recommendations would decrease overall competition (and thus excessive turnover) in the IT labor market. Future work might also examine the incentive effects of organizational changes, such as the introduction of offshoring. Another opportunity for future research is to test our framework on a data set spanning multiple firms. Finally, future work can incorporate human capital and the signaling role of promotions to more fully capture the multiple channels through which a job ladder affects employees' incentives to leave a firm.

While many firms struggle with annual IT turnover rates of 24% or higher, managers have the option to increase pay and benefits to achieve SAS's rate of 6% or adjust the job ladder to lower turnover with a trade-off between promotion incentives and replacement costs. The guidance in this paper helps managers make the right decision for their specific combination of IT needs and industry characteristics.

Supplemental Material

Supplemental material to this paper is available at <http://dx.doi.org/10.1287/isre.2016.0652>.

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CORRECTION

In this article, "Designing Promotion Ladders to Mitigate Turnover of IT Professionals" by Frank MacCrory, Vidyanand Choudhary, and Alain Pinsonneault (first published in *Articles in Advance*, September 1, 2016, *Information Systems Research*, DOI:10.1287/isre.2016.0652), the history line has been corrected.

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